National University of Singapore
School of Computing
CS3244: Machine Learning
Tutorial 8 (1 week)

Neural Networks and Deep Learning

Readings: LFD, dynamic e-Chapter 7

1. **Combining perceptrons.** (Adapted from LFD Exercise 7.1, 7.3, 7.4) On Slide 4 of the lecture notes, we have constructed the XOR function from two simple perceptrons h_1 and h_2. Let's extend this to a more complicated function f shown below.

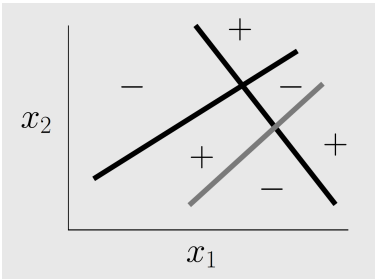


Figure 1: Target function f.

(a) Given the three perceptron components h_1, h_2 and h_3,

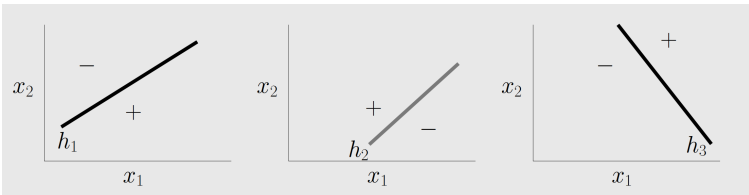


Figure 2: The three perceptron components of f.

verify that

$$f = \overline{h_1}h_2h_3 + h_1\overline{h_2}h_3 + h_1h_2\overline{h_3}.$$

(b) Is there a systematic way to derive the boolean formula for f given its perceptron components? Hint: f given in (a) is of the disjunctive normal form (an OR of ANDs), what do each of the three terms in f correspond to?

(c) Construct the 3-layer multilayer perceptron for f, following the XOR example in the lecture notes.

(d) Using your findings in (b) and (c), provide informal arguments for the following theorem: Any decision function whose ± 1 regions are defined in terms of the regions created by a set of hyperplanes can be implemented by a 3-layer perceptron.

(e) Can the following function in Figure 3 be expressed with a 3-layer perceptron?

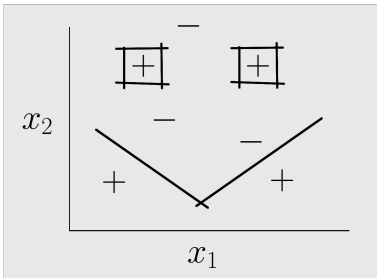
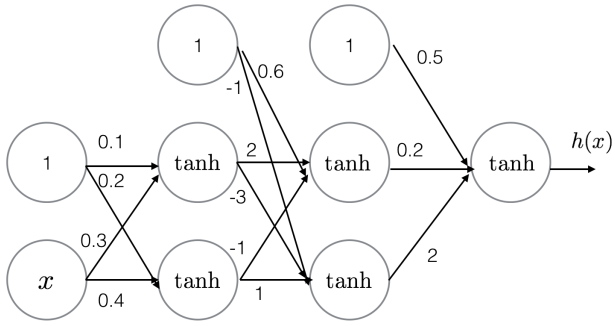


Figure 3: Function for 1(e).

2. **Backpropagation algorithm.** (Adapted from LFD Example 7.1) For the following 3-layer neural network, there is a single data input x and output label y.



The weight matrices are:

$$W^{(1)} = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}; W^{(2)} = \begin{bmatrix} 0.6 & -1 \\ 2 & -3 \\ -1 & 1 \end{bmatrix}; W^{(3)} = \begin{bmatrix} 0.5 \\ 0.2 \\ 2 \end{bmatrix}$$

(a) For the data point $x = 2, y = 1$, perform forward propagation to obtain the final output $\mathbf{x}^{(3)}$.

(b) Perform backpropagation to obtain the errors at each layer (ie. $\boldsymbol{\delta}^{(3)}, \boldsymbol{\delta}^{(2)}, \boldsymbol{\delta}^{(1)}$).

(c) Obtain the partial derivatives of the error with respect to the weights (ie. $\frac{\partial e}{\partial W^{(1)}}, \frac{\partial e}{\partial W^{(2)}}, \frac{\partial e}{\partial W^{(3)}}$).

3. **Regularization with weight elimination.** (Adapted from LFD Exercise 7.11) For soft weight elimination, the regularization term is

$$\Omega(\mathbf{w}) = \sum_{i,j,l} \frac{\left(w_{ij}^{(l)}\right)^2}{1 + \left(w_{ij}^{(l)}\right)^2}$$

Hence, the combined augmented error of the network is

$$E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N} \sum_{i,j,l} \frac{\left(w_{ij}^{(l)}\right)^2}{1 + \left(w_{ij}^{(l)}\right)^2}$$

(a) By taking derivatives with respect to the network weights, show that

$$\frac{\partial E_{\text{aug}}}{\partial w_{ij}^{(l)}} = \frac{\partial E_{\text{in}}}{\partial w_{ij}^{(l)}} + \frac{2\lambda}{N} \frac{w_{ij}^{(l)}}{\left(1 + \left(w_{ij}^{(l)}\right)^2\right)^2}$$

(b) From the above, argue that weight elimination causes small weights to decay faster than larger weights.